Plato’s Pizza Analysis

Plato’s Pizza Analysis: A dynamic, interactive data visualization and statistical analysis project built to explore a year's worth global pizza sales—focusing on revenue trends, sales efficiency, peak ordering hours, order timing, weekday impact, and product performance, inventory management.

Business Objectives

- How much money did we make this year? Can we identify any seasonality in the sales?
- What is the total pizzas sold, and total number of orders placed?
- How are sales distributed across different pizza categories, sizes, and types?
- How do sales trends vary by day and month?
- Which pizzas generate the highest and lowest revenue?
- What is the Average Order Value (AOV) and Average Pizzas per Order, and what do they reveal about customer purchasing behavior?
- How many customers do we have each day? Are there any peak hours?
- How many pizzas are typically in an order? Do we have any bestsellers?

The **Plato’s Pizza Analysis** is a high-performance analytical tool designed to synthesize over 20,000 rows of transactional data into actionable business intelligence. The project aims to identify peak sales periods and quantify the "Opportunity Cost" of weekday late-night closures, highlighting crucial areas of impact like staffing optimization and menu engineering for the client.

Tech Stack

The analysis was conducted using Python and Excel integrated with modern visualization tools:

- 📗 **Microsoft Excel** – Utilized as the primary data integration layer to merge disparate datasets from multiple tables into a single, unified flat file for analysis.

- 🪐 **Jupyter Notebook (Python)** – The core environment for Exploratory Data Analysis (EDA). Used Python libraries (Pandas, Matplotlib, Seaborn) to clean the dataset and generate initial visualizations to answer critical data-driven questions.
- 🧠 **DAX (Data Analysis Expressions)** – Implemented for dynamic KPI modeling, including time-based revenue loss measures.
- 📊 **Power BI Desktop** – Main visualization platform used for report creation and interactive filtering.

Data Source

Source:

The dataset contains detailed records of pizza orders, including `order_id`, `pizza_id`, `quantity`, `order_date`, `order_time`, and `unit_price`. Supplemental tables include a `data_dictionary` for pizza categories (Classic, Veggie, Supreme, Chicken) and sizes (S, M, L, XL, XXL). The data spans a full calendar year to account for seasonality.

Features / Highlights

- **Business Problem** Client often struggle with balancing labor costs against late-night revenue and peak sales hours . Specifically, the client needed to determine if remaining open past 10:00 PM on weekdays (Sunday–Thursday) was financially viable. Additionally, there was a lack of clarity on which pizza categories drove the highest volume versus the highest margin.
- **Goal of the Dashboard** To deliver an interactive visual tool that:
    - Calculates the **Revenue Impact** of late-night weekday operations.
    - Identifies peak order hours to assist in labor scheduling.
    - Segments sales by pizza category and size to optimize inventory.

Key KPIs (Top Summary)

- **Total Revenue:** $817,860
- **Average Order Value:** $38.31
- **Total Pizzas Sold:** 49,574
- **Total Orders:** 21,350

- **Average Pizzas Per Order:** 2.32

Daily & Hourly Sales Trends (Bar Chart & Heatmap)

- **Orders by Day:** Highlights that Friday and Saturday are peak volume days.
- **Orders by Hour:** A peak is observed during lunch (12:00–13:00) and dinner (17:00–18:00), which Figure 1 identifies as the critical windows for maximum staffing.

Sales by Pizza Category (Donut Chart) Visualizes the distribution across Classic, Supreme, Chicken, and Veggie. Classic pizzas typically lead in volume, while Chicken pizzas often contribute higher per-unit margins.

Top & Bottom Sellers (Bar Charts) Ranks the top 5 and bottom 5 pizzas by revenue. This assists in "Menu Engineering"—deciding which items to promote and which to potentially remove.

Size Distribution (100% Stacked Bar/Pie Chart) Breaks down sales by Large (L), Medium (M), and Small (S). Data shows Large pizzas contribute the highest percentage to total revenue, suggesting a consumer preference for value-sized options.

- **Business Impact & Insights**
  - **Operational Efficiency:** By identifying that late-night weekday revenue accounts for a minimal percentage of total impact (calculated via the `% Weekday Revenue Impact` measure), the business can safely reduce hours to save on utility and labor costs.
  - **Staffing Optimization:** The hourly trend analysis allows for "Lean Staffing" during 15:00–16:00 and "Max Staffing" during the 18:00 dinner rush.
  - **Inventory Precision:** Knowing that "Large" and "Classic" are the primary drivers allows for more accurate flour and dough procurement, reducing waste.
  - **Predictive Growth:** Using SAS-based regression, the model can predict future sales spikes based on historical weekday patterns, ensuring the business is never under-prepared.

Business Impact & insights Marketing Optimization

Client can use this dashboard to identify key selling points based on which category's which all pizzas are most ordered. They can also spot least performing pizzas to either remove them from the menu or work to product optimization strategy.

- **Promote High-Revenue Pizzas**
  Focus marketing campaigns and promotional offers on best-selling pizzas to maximize revenue generation and customer satisfaction.
- **Leverage Peak Sales Periods**
  Increase staffing, inventory, and promotional efforts during high-demand days and months to ensure smooth operations and maximize profits.
- **Encourage Larger Orders**
  Introduce combo deals, discounts, or bundle offers to increase the Average Order Value (AOV) and average number of pizzas per order.